

Fin mount — 3D modeling specification

Camera/mic mount for the squash recording iPhone. Clamps a structural glass fin behind the ASB glass back wall; phone periscopes over the wall top edge on the 0.5× ultrawide. Goals, in order: (1) repeatable pose (calibrate once), (2) vibration isolation for the mic, (3) zero protrusion into the court volume, (4) zero risk to the tempered glass.

All dimensions mm. AFF = above finished (court) floor.

1. Confirmed site facts (ASB wall, from photo + manufacturer specs)

Item	Value
Wall / fin / door glass thickness	12 (tempered — no point loads, no hard edge contact)
Wall top edge height	2130 AFF
Fin depth (outward from wall)	~300
Fin height	full wall height — top edge at 2130, coplanar with wall top
Fin anchoring	steel base plate on floor + black patch fittings to wall panel
Door	full height (2130 × 750–900), hinged on the center fins
Wall attachment	fins joined to wall via patch plates — expect a small fin-to-wall gap

2. Governing rules & court constants (WSF Specifications for Squash Courts, 2013/17 ed.)

2.1 Rules that bind the mount design

- **WSF 3.01:** play may be filmed "from above the court or through any of the clear walls" — the over-the-edge camera position is explicitly sanctioned.
- **WSF 3.02:** "No camera or other equipment may project into any part of the court or below the minimum clear height" (clear height falls to 4000 AFF at the back-wall end, WSF 4.04). => **Hard requirement:** every part of the assembly, at worst-case tilt, stays behind the court-side wall face plane. This is constraint 2 in §8 — verify it in CAD, not by eye.
- **WSF 5.03:** on a transparent back wall exactly 2130 high the back wall line is omitted — **the wall top edge itself is the out boundary and is in play**. Balls legitimately strike at and above the edge, so the mount *will* take occasional direct hits: the retention band is mandatory, and the cradle gets a low prow on the court side ahead of the phone's lower edge.
- **WSF 7.01/7.02:** the door sits in the middle third and **opens into the court** — its swing can never reach a mount on the fin outside, and the flush-surface requirement confirms nothing may be added court-side.

2.2 Court geometry constants (also the pipeline's reference numbers)

Measured to finished playing surfaces at 1000 AFF; all heights to the **underside** of markings; markings are 50 wide and are out; line positions accurate to ±2 (WSF 5.01).

Feature	Nominal	Tolerance
Court length	9750	±10
Court width (singles)	6400	±10
Diagonals	11665	±20
Front wall line (out)	4570	—

Feature	Nominal	Tolerance
Service line (front wall)	1780	—
Tin	480	—
Back wall out boundary	2130 (= glass top edge, line omitted)	—
Short line, nearest edge from back wall	4260	—
Service boxes	1600 square, internal	—
Half-court line	equidistant from side walls, short line -> back wall	—

Implication for the pipeline: courts legally differ by up to ± 10 mm and line heights are datumed to the marking's underside — treat WSF nominals as priors and the per-court calibrated homography as truth. The underside of each line is the precise calibration feature.

3. Fin selection

- **Primary: the fin immediately RIGHT of the door (viewed from outside the court).** ~450 mm off court centerline — best horizontal coverage of the far back corner. Door slams ring this fin, but doors slam between rallies, not during play.
- **Fallback: the outer right fin** (panel joint, ~1.8 m off center) if the door hinge / patch hardware fouls the saddle. Same part fits both; only the calibration profile differs.
- Always the same fin on the same court. Mark it (tape dot). One homography + distortion profile per court, keyed by court number.

4. Datum scheme (how the mount self-registers, no adjustments)

Datum	Surface	DOF constrained
A	Fin top edge (horizontal, 2130 AFF)	height (Z), pitch
B	Fin faces (both sides of 12 mm glass)	lateral (Y), yaw, roll
C	Wall glass outer face	setback (X)

Gravity seats A; the jaws close on B; a sprung stop butts C. Drop-on, one thumbscrew, same pose every session. Target repeatability: ≤ 1 mm translation, $\leq 0.3^\circ$ any axis.

5. Measure on site before finalizing CAD (parameters)

Sym	Measurement	Expected	Used in
T_f	Fin thickness incl. any film (calipers)	12.0 ± 0.3	jaw channel
G	Fin-to-wall gap at the junction	2–8	wall-stop boss length
F1	Top patch fitting: distance below top edge	?	jaw notch
F2	Top patch fitting: extent outward from wall face	?	jaw notch / saddle position
F3	Patch fitting plate thickness (standoff off fin face)	~5–10	jaw notch depth

Sym	Measurement	Expected	Used in
E_t	Wall top edge detail (bare polished glass vs. alu cap; chamfer size)	bare	seat profile
P_w×P_h×P_t	Phone + case envelope	model-specific	cradle pocket
E	Ultrawide lens center -> nearest long edge of phone+case	~22	lens setback check
M	Mic port position relative to lens center	model-specific	mic keep-out

6. Assembly overview

- [4] mic scoop (optional clip-on)
- [3] phone cradle (TPU-lined frame, lens corner up, court-side prow)
 - ↓ dovetail on the wedge face + tapered pin
- [2] tilt wedge – A: 40° (16:9) / B: 32° (4:3) <- swappable
 - ↓ dovetail on the mast top + tapered pin
- [1b] mast – dovetail into the saddle cap + tapered pin
- [1a] saddle (straddles the fin top edge at the wall junction)

Fully printed — no purchased hardware (iteration-1 decision: durability traded for zero-BOM). Every joint is a printed dovetail (0.4 clearance, end-stop, Ø5 tapered retaining pin); the clamp is a printed coarse-thread thumbscrew. Saddle and mast are separate parts joined by the same dovetail — a merged part has no support-free print orientation. Relaxed fits throughout — the registration datums (A/B/C) still come from glass contact, so pose repeatability survives sloppy part-to-part fits.

7. Part specifications

7.1 Saddle (PETG)

- **Jaw channel:** width = $T_f + 3.5$ (12 -> **15.5**) between PETG faces; **4 mm TPU 68A pads in 2.5-deep recesses each side (1.5 proud)** -> light slip fit over the glass (~0.25/side), the clamp screw closes the gap; PETG stays ≥ 1.75 off the glass at all times. 68A note: at these pad areas flat-pad stiffness is set by area, not durometer — pose stiffness is unaffected; soft pads squirm in shear, hence the deeper pockets (walls carry shear) and thicker sections. All TPU elements are pocketed. Inner faces parallel ≤ 0.2 . Channel length along fin: **130**. Jaw depth below top edge: **80** fixed jaw (wall side may shorten to clear F1/F2/F3 — model the patch-fitting notch parametrically).
- **Cap** (bridges the edge): 15 thick above the edge; underside carries a **3 × 12 × 110 TPU 68A seat strip** in a 2.5-deep recess, 0.5 proud (datum A — glass edge never touches PETG).
- **Wall stop** (datum C): two bosses at the wall end, faced with 3 mm TPU 68A in 2-deep recesses (1 proud), contacting the wall outer face; boss length = $G + 2$ (parametric); vertical spread ≥ 60 for a stable couple.
- **Clamp (fully printed):** coarse printed thread — **M12 × 2.5 trapezoid-profile thumbscrew** (PETG, integrated Ø32 notched knob and Ø8 ball tip) threading directly into a printed nut thread in the outer-jaw boss (0.4 radial clearance). A printed **pad cap** (Ø18, ball-socket snap-fit over the tip, 2.5 mm 68A face disc in a pocket) lands flat on the glass and doesn't rotate while tightening. Advancing the screw presses the glass onto the fixed jaw's bonded pad (datum B). Snug finger-tight only ($\leq \sim 50$ N) — the soft 68A face conforms and makes over-tightening obvious by feel; the screw holds position, the geometry holds pose.
- Edge fillets ≥ 2 everywhere near glass; no printed corner may touch a glass edge.

7.2 Mast (PETG, joined to the saddle by dovetail + pin)

- Column $\geq 30 \times 45$ rising from the cap's wall end so the cradle lands over the wall glass. Base carries a male rail underneath that slides into a slot in the cap top (pinned); base width = column width so the part lies flat on its

side for printing.

- Top face carries the **wedge dovetail**: male rail (root 20, crest 26, height 6, ~63° flanks — prints support-free), hard end-stop, axis parallel to the wall. Wedge is retained by a Ø5 printed tapered pin dropped through the wedge into the column (friction fit).

7.3 Tilt wedges (PETG) — print both

Wedge	Optical axis pitch	Capture mode	Frame top hits	Engrave
A	40° below horizontal ±0.5°	ultrawide 16:9 (stock)	front-wall service line (1780)	UW-16:9-40
B	32° below horizontal ±0.5°	ultrawide 4:3 (full sensor, custom AVFoundation)	front-wall out line (4570)	UW-4:3-32

Residual angle error is absorbed by calibration — repeatability matters, accuracy doesn't. Each wedge: female dovetail underneath (mast rail + 0.4 clearance), and its **own male rail + end-stop on the inclined face** for the cradle — same profile, same tapered-pin retention.

7.4 Phone cradle (PETG body, TPU liner)

- Pocket = phone+case envelope + 0.4 clearance per side; **liner is a single drop-in TPU insert (1.5 wall)** covering every contact face, keyed into the PETG pocket with shallow dovetail ribs — the phone never touches PETG (grip + structure-borne vibration decoupling), and the liner is replaceable if it wears.
- Landscape, **ultrawide lens corner at the top, facing the court.**
- Corner window for the whole lens cluster + mic: no material within a **Ø40 clear zone** around the mic port (parameter M), none within the FOV frustum (§8).
- Retention: insertion from the top, retention lip + **band grooves (mandatory — the wall top edge is in play, WSF 5.03, and the mount will take occasional ball strikes)**. Bands are **printed TPU 68A rings** (ID ~48, 4 × 2.5 section — 68A stretches far enough). No rigid latch over the top edge — it would sit in the FOV.
- Mounts to the wedge via a **female dovetail boss on the back** (slot along the wall direction, end-stop, tapered pin).
- **Prow**: low 45° chamfered ridge on the cradle's court side ahead of the phone's lower edge, below the FOV frustum, to shed direct ball strikes.
- **USB-C slot** 14 × 8 at the phone's lower edge + cable channel with strain-relief boss down the mast (match recordings need wall power).

7.5 Mic scoop (optional, TPU)

Clip-on horn around the mic port aimed down-court. Separate part, zero contact with the lens window, removable for A/B testing. Any scoop change => audio threshold retune.

8. Optical placement (drives mast + wedge geometry)

Parameter	Value	Derivation
Lens center height	2220 AFF (90 above top edge)	see setback coupling below
Lens center setback	18 outboard of the court-side wall face (= 6 outboard of the wall's outer face)	edge-occlusion + phone-lean limits
Optical axis yaw	0° (⊥ back wall) ±0.5°	datum B
Optical axis pitch	wedge A/B (§7.3)	
Cradle lateral position	phone center within ±20 of the fin midplane	mass over the clamp

Governing constraints (re-verify in CAD if any input changes): 1. **Edge occlusion**: bottom of frame is $\sim 76.5^\circ$ below horizontal (both modes). Require $h / s \geq \tan 78.5^\circ = 4.92$ (2° margin), h = lens height above edge, s = lens setback from the court-side face. $90 / 18 = 5.0$ [ok] 2. **Court-plane clearance (WSF 3.02 — mandatory)**: at 40° tilt the phone's upper long edge leans court-side of the lens by $E \cdot \sin 40^\circ$ (~ 12.9 for $E = 22$). Require $s - E \cdot \sin(\text{tilt}) \geq 3.18 - 12.9 = 5.1$ [ok] — **no part of the assembly may cross the court-side face plane**. Check the prow, band, and cable too. 3. **FOV keep-out frustum** from the lens center: $\pm 55^\circ$ horizontal, $+15^\circ$ to -78° vertical (covers both wedges). Nothing — cradle, band, prow, scoop, cable — inside it.

9. Materials, hardware, print settings

Part	Material	Orientation / notes
Saddle	PETG (ASA if the gallery gets hot sun)	upside-down (cap on plate, jaws opening up) — datum pockets print crisp; boss chamfers + teardrop make it support-free
Mast, wedges	PETG	lying on the side — every feature is a Y-extrusion, support-free
Cradle	PETG	back boss down — the ONE supported part (under the plate wings)
Clamp screw, pad cap, pins	PETG	screw vertical, knob down
Pads, liner, bands, scoop, swivel face	TPU 68A	print flat; glued (CA) into recesses

0.2 layers, 4 perimeters, $\geq 40\%$ gyroid on saddle+mast. **No PLA anywhere** — creep under sustained clamp load is pose drift, which defeats the calibrate-once design. **STLs are exported already posed for printing** — no rotation in the slicer.

BOM: none. Iteration 1 is fully printed — printed M12×2.5 clamp thread, printed dovetails + tapered pins, printed 68A bands. Durability is explicitly traded away; if a printed thread or pin wears out, reprint it (minutes, not a hardware-store trip).

BOM: M5 brass heat-set inserts ×7 · M5×25 SS thumbscrew ×1 (clamp) · M5×16 SS ×5 (mast bolts + wedge lock) · Ø4×10 SS dowel pins ×2 · Ø18 swivel pad ×1 · silicone band ×2.

10. Install & verification

Install: hook cap over fin edge at the junction -> slide until wall stops touch -> finger-tighten thumbscrew -> seat phone -> band. Target < 30 s.

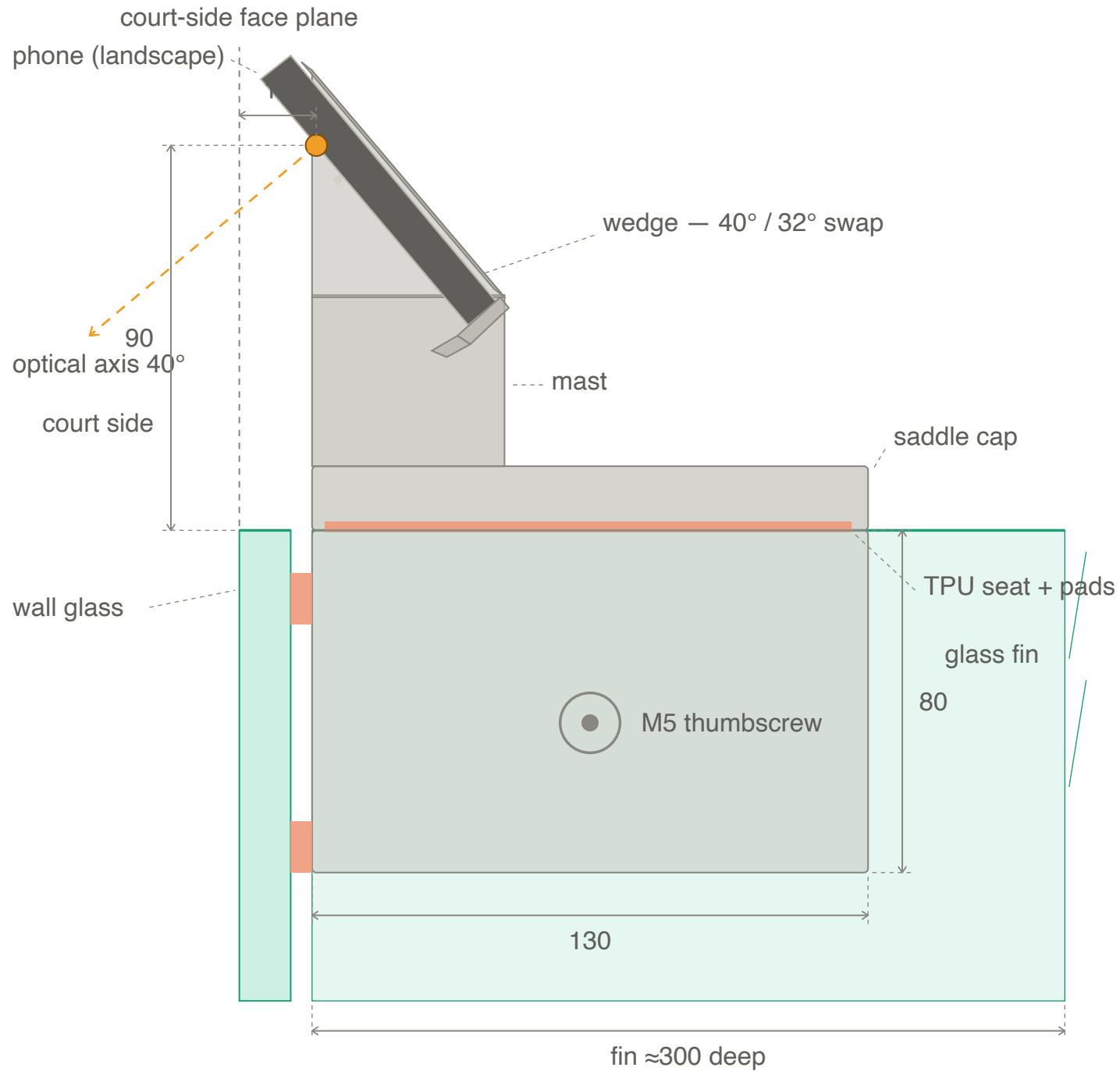
Acceptance tests before trusting data: 1. **Repeatability**: 10× full remount; court-line homography corner residuals ≤ 2 px (calibrate against marking **undersides** — the WSF datum, §2.2). 2. **Ball strike / door slam**: record while striking the wall near the top edge and slamming the door; verify no frame shift > 1 px and that structure-borne transients don't trigger the classifier. 3. **FOV**: both wedges — floor visible to ≤ 0.6 m of the back wall; frame top at service line (A) / out line (B); wall edge not in frame. 4. Then: one-time ultrawide distortion calibration -> homography -> **audio threshold retune** (new mic position invalidates current levels) -> re-run the ground-truth eval.

11. Open items

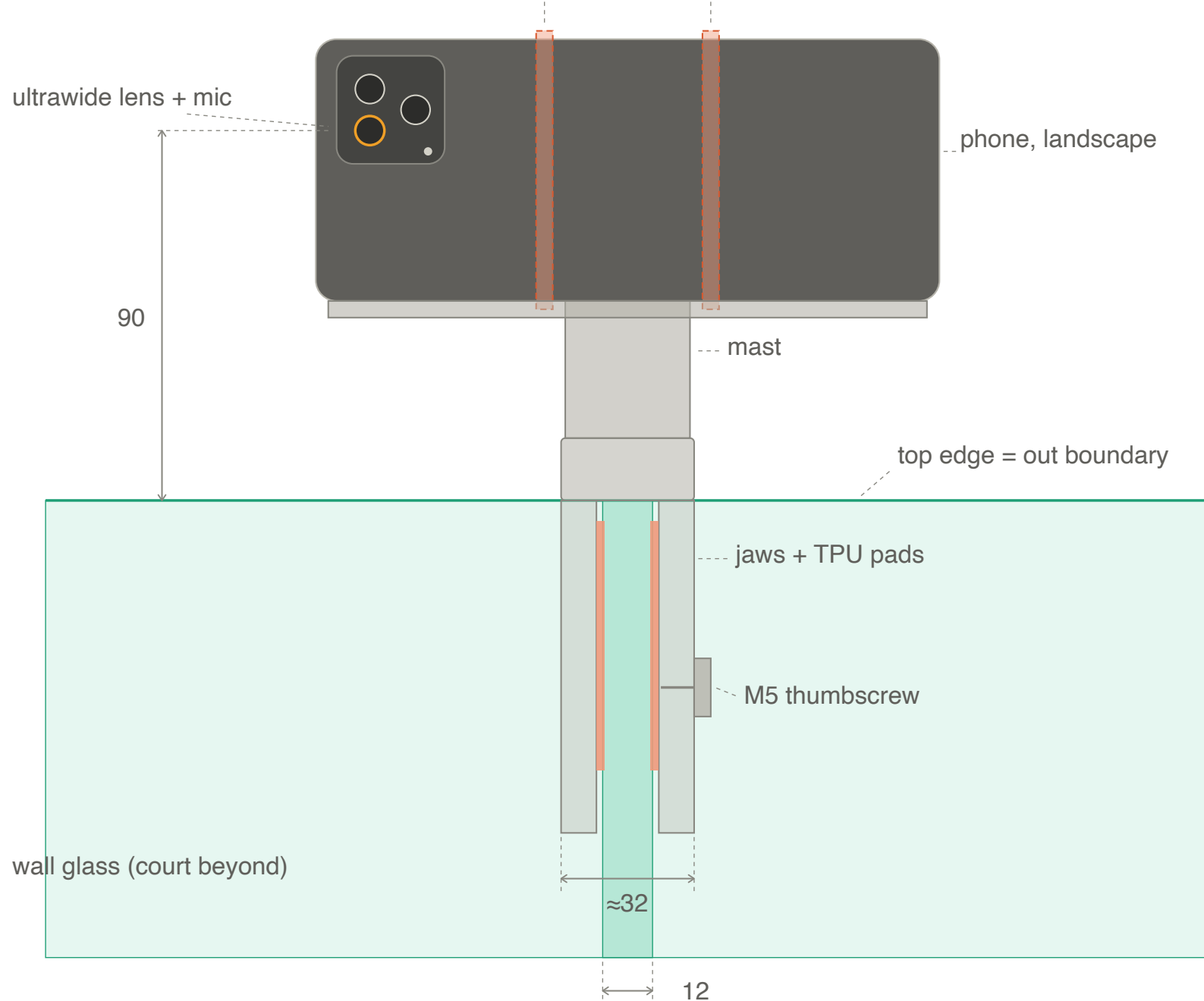
- [] Site measurements table (§5)
- [] Confirm door-fin patch/hinge hardware clears the saddle, else fall back to outer fin
- [] Phone model + case -> cradle pocket, E, M
- [] 4:3 full-sensor capture path in the pipeline (custom AVCaptureSession) before printing wedge B matters

- [x] Cross-check `court_model.py` constants against §2.2 (underside datum, 50 mm line width) — done 2026-07-20: all ft constants land on the WSF datum edges within ~4 mm (inside the ± 10 mm build tolerance); datum convention documented in `court_model.py`. Follow-up same day: wizard snap landmarks re-datumed to paint CENTERLINES (`LINE_HALF_WIDTH_FT` shift), because the snap refiner's RANSAC+PCA fit lands mid-stripe — the edge datum left every snapped landmark ~25 mm biased (~35 mm diagonal at box corners), vs ~0.26 in/px at the short line. Zone/judge constants stay WSF edge datums; labels now say "middle of the line's width" so the no-snap tap fallback matches.

Side elevation — looking along the wall (2 px = 1 mm)



Front elevation — from outside the court (2 px = 1 mm)
silicone bands x2



Plan view — from above (2 px = 1 mm)

